



Artemisia argyi attenuates airway inflammation in ovalbumin-induced asthmatic animals



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ABSTRACT

Ethnopharmacological relevance: *Artemisia argyi* is a traditional herbal medicine in Korea and commonly called as mugwort. It is traditionally used as food source and tea to control abdominal pain, dysmenorrhea, uterine hemorrhage, and inflammation.

Aim of the study: We investigated the effects of *A. argyi* (TOTAL) and dehydromatricarin A (DA), its active component on ovalbumin (OVA)-induced allergic asthma.

Materials and methods: The animals were sensitized on day 0 and 14 by intraperitoneal injection of OVA with aluminum hydroxide. On day 21, 22 and 23 after the initial sensitization, the animals received an airway challenge with OVA for 1 h using an ultrasonic nebulizer. TOTAL (50 and 100 mg/kg) or DA (10 and 20 mg/kg) were administered to mice by oral gavage once daily from day 18–23. Airway hyperresponsiveness (AHR) was measured 24 h after final OVA challenge.

Result: TOTAL and DA treated animals reduced inflammatory cell counts, cytokines and AHR in asthmatic animals, which was accompanied with inflammatory cell accumulation and mucus hypersecretion. Furthermore, TOTAL and DA significantly declined Erk phosphorylation and the expression of MMP-9 in asthmatic animals.

Conclusion: In conclusion, we indicate that Total and DA suppress allergic inflammatory responses caused by OVA challenge. It was considered that *A. argyi* has a potential for treating allergic asthma.

1. Introduction

Inflammation is an important defense system against harmful stimuli, however; excess responses causes various diseases such as allergic asthma, arthritis and dermatitis via destruction of normal physiological condition (Kim et al., 2012). Of inflammatory diseases, allergic asthma is induced by exposure to specific allergens including pollens, chemicals and air pollutants and featured as airway inflammation, mucus hypersecretion and airway hyperresponsiveness (AHR), resulting in aggravating normal lung functions (Xiong et al., 2012; Lee et al., 2012a, 2012b; Zhao et al., 2017). Allergic inflammation is mainly characterized by eosinophil infiltration and production of allergen-specific immunoglobulin (Ig) E (Lee et al., 2012a, 2012b). Eosinophil infiltration is controlled by Th2 cytokines and IgE and plays a crucial role in the manifestation of asthmatic clinical signs, such as mucus hypersecretion and airway hyperresponsiveness. Excess eosinophil

infiltration is also associated with the activation of inflammatory signaling pathways with the release of cytotoxic granule proteins and enzymes, including matrix metalloproteinases (MMPs) (Choi et al., 2012; Kim et al., 2016; Moore et al., 2011).

MMPs play a key role in maintaining normal physiological functions due to their capacity to degrade structural materials, such as elastin and collagens (Mehra et al., 2010; Cui et al., 2017). These functions of MMPs are closely associated with airway remodeling in the development of allergic asthma and are involved in the production of cytokines, chemokines, and growth factors (Grzela et al., 2016). MMP-9 is associated with inflammatory cell accumulation, remodeling processes, and healing of injured tissue (Lee et al., 2012a, 2012b; Bruschi et al., 2014). MMP-9 production can be induced by extracellular signal-regulated kinase (Erk) signaling (Deng et al., 2013; Li et al., 2010). Erks also play an important role in eosinophil differentiation and activation, and are closely involved in development and

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aggravation of allergic asthma via modulation of interleukin (IL)-5 production (Alam and Gorska, 2011; Liang et al., 2016).

The genus *Artemisia*, commonly called mugwort, has been widely used as a tea, spice, and food ingredient in East Asia (Kim et al., 2015a, 2015b). The mugwort is a Korean native, *A. argyi*, which is frequently used as a traditional medicine for patients with abdominal pain, dysmenorrhea, uterine hemorrhage, and inflammation (Kim et al., 2015a, 2015b; Yun et al., 2016). According to previous reports (Kim et al., 2015a, 2015b; Lee et al., 2015; Yao et al., 2016), mugwort possesses antioxidant and anti-inflammatory properties and contains high amounts of vitamin C and total phenolic compounds. However, there have been no studies on the effect of *A. argyi* against allergic asthma.

Thus, we have explored the therapeutic effects of *A. argyi* (TOTAL) constituent, dehydromatricarin A (DA), against airway inflammation, using a murine asthma model induced by ovalbumin (OVA). We have also investigated the expression and production of allergic, asthma-related inflammatory mediators to elucidate a possible mechanism of *A. argyi*.

2. Materials and methods

2.1. Plant material

Artemisia argyi was collected from a farm in Eumseong-gun, South Korea, in May of 2016 (National Institute of Horticultural and Herbal Science, NIHHS). After harvesting, the aerial parts were immediately freeze-dried and stored at -40°C until analysis. A voucher specimen (KRIBB 0076330) was deposited at the Plant Extract Bank of Korea Research Institute of Bioscience and Biotechnology (KRIBB) in Daejeon, south Korea.

2.2. Extraction and isolation

First, *A. argyi* was washed with clean, sterile water. The raw material was cut into small pieces, and the chopped leaves were mixed with three volumes of methanol. The mixture was sonicated three times for 1 h. Evaporation of solvent under reduced pressure yielded the crude extracts (94.0 g, 10.5% yield), which were used for analysis of biological activities and ultra-performance liquid chromatography quadrupole time of flight mass spectrometry (UPLC-QToF-MS).

2.3. UPLC-QToF-MS analysis

Profiling of TOTAL was conducted on an ACQUITY UPLCTM system (Waters Corporation, Milford, MA, USA) equipped with a binary solvent delivery manager and a sample manager. This was coupled to a Micromass Q-TOF PremierTM mass spectrometer (Waters Corporation) with an electrospray ionization (ESI) interface and MassLynx V4.1 software. Chromatographic separation was conducted on an ACQUITY BEH C₁₈ chromatography column (2.1 × 100 mm, 1.7 μm). The column was maintained at 35 °C, and the mobile phases A and B were 0.1% formic acid in water and 0.1% formic acid in acetonitrile, respectively. The gradient elution program was as follows: 0.0–1.0 min, 15% B; 1.0–6.0 min, 15–20% B; 6.0–14.0 min, 20–55% B; 14.0–16.3 min, 55–100% B; wash for 1.5 min with 100% B; and then a 2.0 min recycle time. The injection volume and flow rate were 5.0 μL and 0.4 mL/min, respectively. The mass spectrometer was operated in negative ion mode. Leucine-enkephalin and N₂ were used as reference compound (m/z 556.2771 in the positive mode) and desolvation gas, respectively. The flow rate was 500 L/h, desolvation temperature was 350 °C, source temperature was 100 °C, capillary voltage was 2700 V, and cone voltage was 27 V. The data were collected for each test sample from 200 Da to 1500 Da with a 0.25-s scan time and 0.01-s interscan delay over a 25-min analysis time.

2.4. Experimental procedure

Female BALB/c mice (20–25 g, 6 weeks old, Samtako Co., Osan, Korea) were maintained in groups ($n = 8$ per group) in a controlled environment (temperature $22 \pm 2^{\circ}\text{C}$ and humidity $55 \pm 5\%$). The experimental design was approved by the Institutional Animal Care and Use Committee of Chonnam National University. The animals were randomly divided into 7 groups: normal control group (NC; no treatment and phosphate-buffered saline (PBS) challenge), ovalbumin (OVA) challenge group (OVA; no treatment, sensitization, and OVA challenge), dexamethasone group (DEX; 3 mg/kg dexamethasone, sensitization, and OVA challenge), *A. argyi* groups (TOTAL50 and TOTAL100; 50 and 100 mg/kg *A. argyi*, respectively, sensitization, and OVA challenge), and dehydromatricarin A groups (DA10 and DA20; 10 and 20 mg/kg dehydromatricarin A, respectively, sensitization, and OVA challenge). The dose of *A. argyi* was determined according to preliminary study. In preliminary study, *A. argyi* was administered to mice at doses of 100 and 200 mg/kg body weights. The experimental procedure was the same with present study. BALF cell counts were observed the significant decreases in both 100 and 200 mg/kg treated mice. However, 30 mg/kg treated mice was more effective than 60 mg/kg treated mice. Based on these results, we determined 100 mg/kg, an as experimental dose. OVA induced allergic asthma model were designed as previously described (Lee et al., 2012a, 2012b; Shin et al., 2015). To induce allergic inflammation, sensitization was performed via intraperitoneal injection of OVA (20 μg) mixed with aluminum hydroxide (2 mg) on days 0 and 14. The animals received nebulized 1% (w/v) OVA solution for 1 h from day 21–23, using a nebulizer (Omron, Tokyo, Japan). Drug administration was performed 1 h before the OVA challenge from day 18–23. Airway hyperresponsiveness (AHR) was measured using whole-body plethysmography (Allmedicus, Seoul, Korea) on day 24. AHR was measured following methacholine challenge (doses of 0, 5, 10, and 15 mg/mL in PBS) for 3 min. Results were recorded as the dimensionless parameter, enhanced pause (Penh).

2.5. BALF analysis

Twenty-four hours after AHR, bronchoalveolar lavage fluids (BALF) were collected as previously described (Shin et al., 2015). Briefly, a tracheostomy was performed under slight anesthesia and inserted endotracheal tube into trachea. Ice-cold PBS (0.7 mL) was infused into the lung and withdrawn through tracheal cannulation two times (total volume 1.4 mL). To count inflammatory cells, 100 μL of BALF was centrifuged onto slides using a Cytospin (Hanil, Seoul, Korea) and the slide were dried and stained with Diff-Quik[®] reagent (IMEB Inc., Deerfield, IL). Serum was collected by centrifugation of blood obtained from the caudal vena cava. Quantitative analyses of IgE (Biolegends, San Diego, CA) and Th2 cytokines (R&D System, Minneapolis, MN) were performed using ELISA kits with serum and BALF, respectively. Optical absorbance was measured using a spectrophotometer (Bio-Rad, Hercules, CA, USA).

2.6. Immunoblotting

To perform immunoblotting, lung tissues were homogenized with a tissue lysis buffer (Sigma-Aldrich) and its protein concentration was assayed with Bradford solution (Bio-Rad). Immunoblotting was performed as previously described (Shin et al., 2015). Equal amounts of protein (30 μg) were resolved by 10% SDS-polyacrylamide gel electrophoresis and transferred to nitrocellulose membrane. The membrane was incubated with 5% skim milk, followed by an overnight incubation with the primary antibodies. Anti-β-actin, anti-Erk, anti-phospho-Erk (all from Cell Signaling Technology, Danver, MA), and anti-MMP-9 (Abcam, Cambridge, MA) were used as primary antibodies. The membranes were washed for three times with Tris-buffered

saline containing Tween 20 (TBST), and then incubated with secondary antibody (Jackson Immuno Research PA, USA) for 30 min. The membranes were washed three times with TBST and then developed using an enhanced chemiluminescence agents (R & D System). Protein expression levels were evaluated by Chemi-Doc (Bio-Rad).

2.7. Gelatin zymography

To determine MMP-9 activity, gelatin zymography was conducted according to a previous report (Shin et al., 2014). Briefly, the zymogram gels consisted of 10% SDS-PAGE containing 1% gelatin as an MMP substrate. The gels were washed in 2.5% Triton X-100 for 1 h to remove SDS and then incubated at 37 °C for 16 h in developing buffer (1 M Tris-HCL, pH 7.5 with CaCl₂). The gels were stained with 25% methanol/8%acetic acid containing Coomassie brilliant blue. The gelatinase activity was visualized as whit bands on a blue background and represented areas of proteolysis using Chemi-Doc (Bio-Rad).

2.8. Lung tissue histopathology

Lung tissues were fixed in 10% (v/v) neutral buffered formalin, embedded in paraffin, sectioned to 4-μm thickness, and then stained with hematoxylin and eosin (H & E) and periodic acid–Schiff (PAS) to visualize inflammation and mucus production, respectively.

To determine protein expression in lung tissue, we conducted immunohistochemistry (IHC), according to a previous study (Shin et al., 2014). The paraffin-embedded sections were deparaffinized, dehydrated, washed using PBS with 0.3% Triton X-100 and incubated for 10 min at room temperature with 10% goat serum to block non-specific staining. Subsequently, slides were incubated overnight at 4 °C with primary anti-mouse MMP-9 (Cell Signaling). After removing the primary antibody, the sections were washed and then incubated with biotinylated secondary antibody at 37 °C for 1 h, followed by incubation with avidin-biotin-peroxidase complex (Vector Laboratories, CA, USA) for 1 h room temperature. The excess complex was removed and sections were washed with PBS prior to incubation in 0.05% diaminobenzidine (Millipore, MA, USA) for a further 10 min. The sections were counterstained, rinsed with PBS to terminated the reaction, and protected with coverslips prior to microscopic examination.

2.9. Statistical analysis

The data are expressed as mean ± standard deviation. Statistical analysis was performed using an analysis of variance (ANOVA), followed by Dunnett's multiple comparison testing. A value of $P < 0.05$ indicated a significant difference.

3. Result

3.1. Identification of active compounds

UPLC-PDA-QTOF-MS analyses were carried out using a C18 column with a linear gradient of acetonitrile/water (Fig. 1). All peaks were characterized using Mass. The m/z values (mass-to-charge ratio) of molecular ions $[M+H]^+$ of compounds 1–13 were 517, 517, 517, 287, 317, 679, 331, 303, 361, 345, 375, 549, and 549, respectively (Supplementary material). Table 1 shows the retention times, UV–Vis absorption maxima, mass spectral data of molecular ions of these compounds. Compounds 1–13 were identified as 1,3-*O*-dicafeoylquinic acid (1), dicafeoylquinic acid (2), 1,4-*O*-dicafeoylquinic acid (3), luteolin (4), isorhamnetin (5), tricafeoylquinic acid (6), jaceosidin (7), dehydromatricarin (8), centaureidin (9), eupatilin (10), casticin (11), artanomaloide (12), and 8-acetylarteminolide (13) (Lee et al., 2000; Wang et al., 2012; Melguizo-Melguizo et al., 2014; Kim et al., 2015a, 2015b; Zhang et al., 2016).

3.2. *Artemisia argyi* reduces inflammatory cell count in BALF of asthmatic animals

Asthmatic animals showed significant elevation of inflammatory cell counts in BALF in comparison to the counts of the normal controls. Specifically, asthmatic animals clearly increased eosinophil counts in comparison with those of normal controls. However, *Artemisia argyi* and dehydromatricarin A treated animals had lower eosinophil counts than the asthmatic animals did (Fig. 2). These reductions were observed in dose-dependent manner.

3.3. *Artemisia argyi* decreases the AHR of asthmatic animals

Asthmatic animals were more increased AHR with increasing methacholine concentration than normal controls. However, dexamethasone treated animals markedly declined AHR in comparison to asthmatic animals. Similarly, *Artemisia argyi* and dehydromatricarin A treated animals were lower AHR than asthmatic animals, which were detected in dose-dependent manner (Fig. 3)

3.4. *Artemisia argyi* reduces Th2 cytokines and IgE of asthmatic animals

Asthmatic animals showed the elevations of IL-4, IL-5, and IL-13 in BALF compared to those of normal controls. Dexamethasone-treated animals showed reduced Th2 cytokines compared with those of asthmatic animals. Similarly, *Artemisia argyi* and dehydromatricarin A treated animals had significantly lower levels of IL-4, IL-5, and IL-13 than asthmatic animals did, which were observed in dose-dependent manner (Fig. 4A, B, and C). Particularly, the production of Th2 cytokines with 20 mg/kg DA (DA20) was similar to that of dexamethasone-treated animals. OVA-specific IgE was elevated in asthmatic

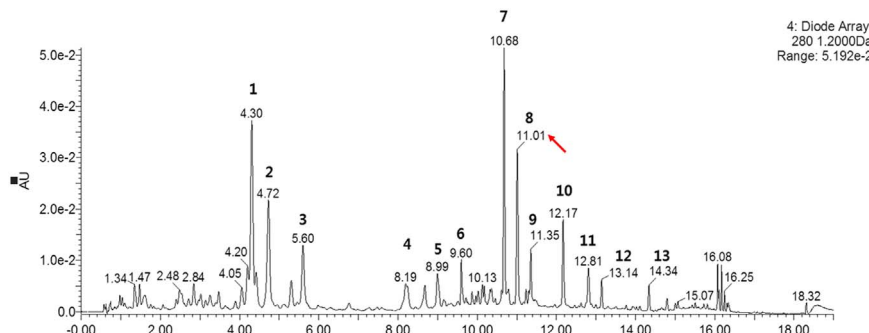


Fig. 1. Representative UPLC-QTOF-MS chromatograms of the methanol extracts from *Artemisia argyi*. Peak numbers refer to Table 1.

Table 1Identification of phytochemical metabolites by UPLC-QTOF-MS from *A. argyi*.

Comp.	Identity	t _R (min)	Calculated [M+H] ⁺	Detected precursor ion [M+H] ⁺	Error (ppm)	Molecular formula	UV
1	1,3-O-dicaffeoylquinic acid	4.31	517.1346	517.1349	0.6	C ₂₅ H ₂₄ O ₁₂	326, 245
2	Dicaffeoylquinic acid	4.73	517.1346	517.1343	−0.6	C ₂₅ H ₂₄ O ₁₂	327, 245
3	1,4-O-dicaffeoylquinic acid	5.60	517.1346	517.1345	−0.2	C ₂₅ H ₂₄ O ₁₂	327, 245
4	Luteolin	8.70	287.0556	287.0534	−7.7	C ₁₅ H ₁₀ O ₆	348, 250
5	Isorhamnetin	9.05	317.0661	317.0629	−10.1	C ₁₆ H ₁₂ O ₇	346, 214
6	Tricaffeoylquinic acid	9.64	679.1663	679.1669	0.9	C ₃₄ H ₃₀ O ₁₅	324, 219
7	Jaceosidin	10.68	331.0818	331.0822	1.2	C ₁₇ H ₁₄ O ₇	344, 254
8	Dehydromatricarin	11.01	303.1232	303.1248	5.3	C ₁₇ H ₁₈ O ₅	262
9	Centaureidin	11.35	361.0923	361.0919	−1.1	C ₁₈ H ₁₆ O ₈	352, 254
10	Eupatilin	12.17	345.0974	345.0965	−2.6	C ₁₈ H ₁₆ O ₇	342, 254
11	Casticin	12.81	375.1080	375.1082	0.5	C ₁₉ H ₁₈ O ₈	350, 254
12	Artanomaloid	13.14	549.2488	549.2537	8.9	C ₃₂ H ₃₆ O ₈	258
13	8-Acetylarteminolide	14.34	549.2488	549.2542	9.8	C ₃₂ H ₃₆ O ₈	258

animals than in normal controls (Fig. 4D). However, *Artemisia argyi* and dehydromatricarin A treatment dose-dependently lowered than OVA specific IgE in asthmatic animals.

3.5. *Artemisia argyi* attenuates the expression and activity of MMP-9 in asthmatic animals

Asthmatic animals showed noticeable elevation of MMP-9 expression in comparison to normal controls. By contrast, *Artemisia argyi* and dehydromatricarin A treated animals showed a significant reduction in dose-dependent manner compared to asthmatic animals (Figs. 5A and 6A). In zymography, MMP-9 activity was higher in asthmatic animals than in normal controls. However, *Artemisia argyi* and dehydromatricarin A treated animals exhibited an dose-dependent reduction in MMP-9 activity compared to that of asthmatic animals (Fig. 5B).

3.6. *Artemisia argyi* reduces Erk phosphorylation in asthmatic animals

Erk phosphorylation was higher in asthmatic animals than in normal controls (Fig. 6B). Dexamethasone treatment markedly reduced Erk phosphorylation in asthmatic mice. In addition, *Artemisia argyi* and dehydromatricarin A treated animals exhibited the reductions in Erk phosphorylation in comparison to that in asthmatic animals, which were detected in dose-dependent manner.

3.7. *Artemisia argyi* attenuates inflammatory cell accumulation and mucus production in asthmatic animals

Asthmatic animals exhibited extensive inflammatory cell accumulation in perivascular and peribronchial lesions (Fig. 7A) and mucus

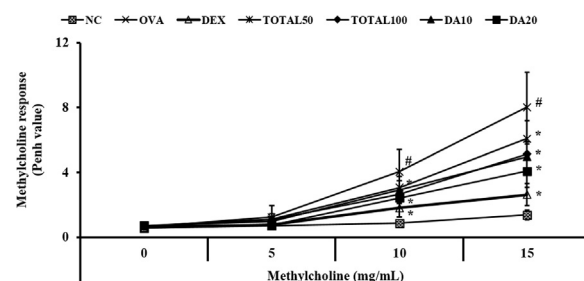


Fig. 3. *Artemisia argyi* reduced airway hyperresponsiveness. NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. * $P < 0.05$ vs NC; * $P < 0.05$ vs OVA.

hypersecretion in airways (Fig. 7B) compared to those of normal control mice. By contrast, *Artemisia argyi* and dehydromatricarin A treatment reduced inflammatory cell accumulation and mucus secretion in asthmatic animals.

4. Discussion

Asthma is considered a threatening health issue that lowers the quality of human life, and its prevalence has gradually increased. In this study, we explored the therapeutic effect of *Artemisia argyi* and its constituent, dehydromatricarin A on allergic responses in a murine asthma model caused by OVA challenge. *Artemisia argyi* and dehydromatricarin A treatments reduced AHR, inflammatory cell counts, OVA-specific IgE, and Th2 cytokines in asthmatic animals, which were consistent with the results of histological examination. Furthermore,

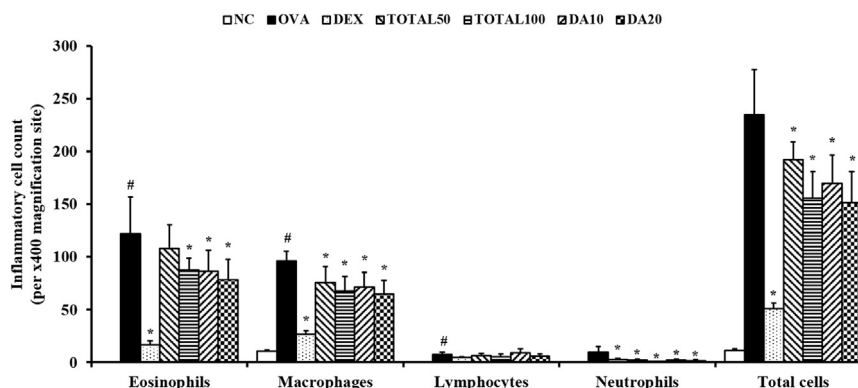


Fig. 2. *Artemisia argyi* declined inflammatory cell counts in the BALF. NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. * $P < 0.05$ vs NC; * $P < 0.05$ vs OVA.

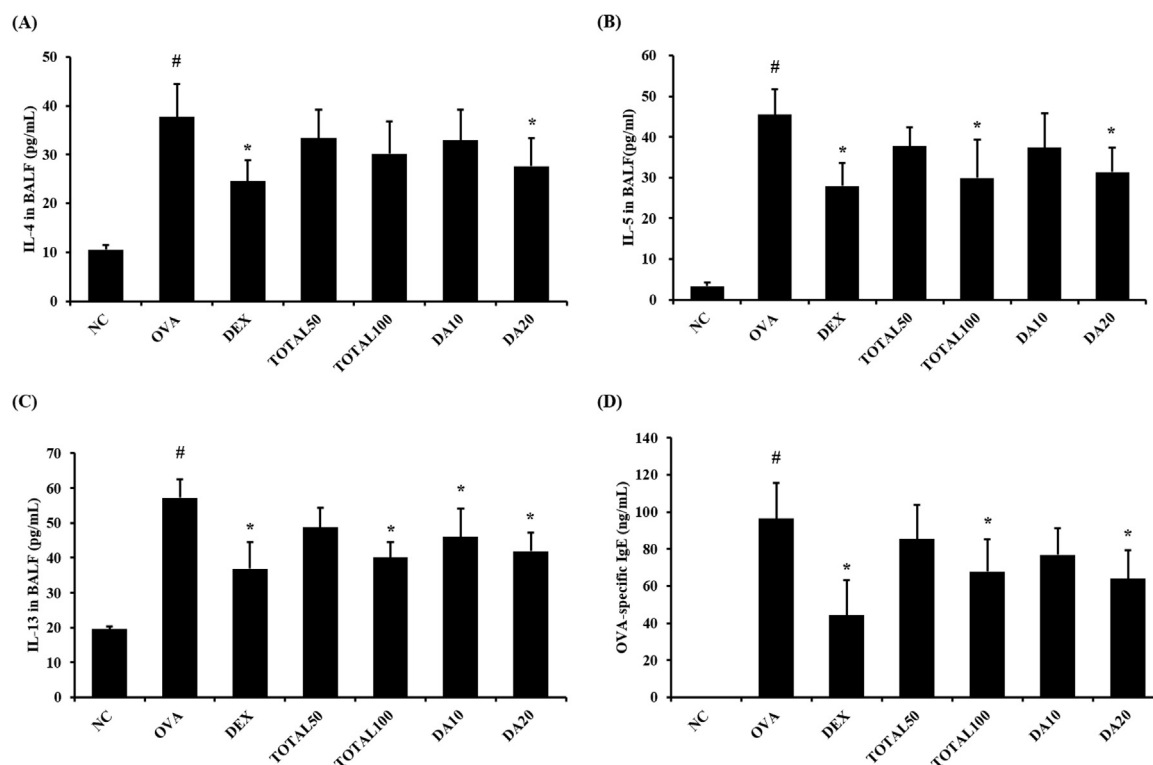


Fig. 4. *Artemisia argyi* decreased level of allergen specific Ig E and Th2 cytokines. (A) IL-4, (B) IL-5, (C) IL-13, (D) OVA-specific IgE. NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. * $P < 0.05$ vs NC; * $P < 0.05$ vs OVA.

Artemisia argyi and dehydromatricarin A treatments significantly reduced Erk phosphorylation and MMP-9 expression induced by OVA challenge, which were observed in dose-dependent manner.

The high potency of the *Artemisia argyi* indicated the presence of compounds that may be therapeutic against various diseases (Kim et al., 2015a, 2015b). We were very encouraged by these previous results with sesquiterpene lactones focused our attention on a more thought interrogation of inflammatory activity with NO and IL-6 (Kim et al., 2015a, 2015b; Zhang et al., 2016). Therefore, the major compounds of *Artemisia argyi* were looking forward to exhibit inflammation inhibitory activities. The discussion of allergic lung inflammation will focus on dehydromatricarin A (DA, **8**) which emerged to be very high concentrations because they appeared as principal peaks in chromatogram.

AHR is considered an important clinical sign of asthma (Park et al., 2016; Jang et al., 2016; Hung et al., 2013). Inflammatory cell activation, including activation of eosinophils, neutrophils, and macrophages, is involved in the occurrence of AHR. Particularly, eosinophil activation is closely associated with AHR because eosinophils act as pathogenic players to develop and exacerbate allergic responses (Kim et al., 2016). Eosinophils are controlled by Th2 cells, and they accumulate in the airway lumen and lung tissue (Jung et al., 2016). They not only produce pro-inflammatory cytokines, but also secrete lipid mediators and cationic proteins via degranulation, resulting in allergic responses (Sun et al., 2015). Th2 cytokines, such as IL-4, IL-5, and IL-13, are associated with the recruitment, maturation, and activation of eosinophils in the development of asthma, and eventually induce AHR and mucus overproduction (de Groot et al., 2015; Jeon et al., 2015;

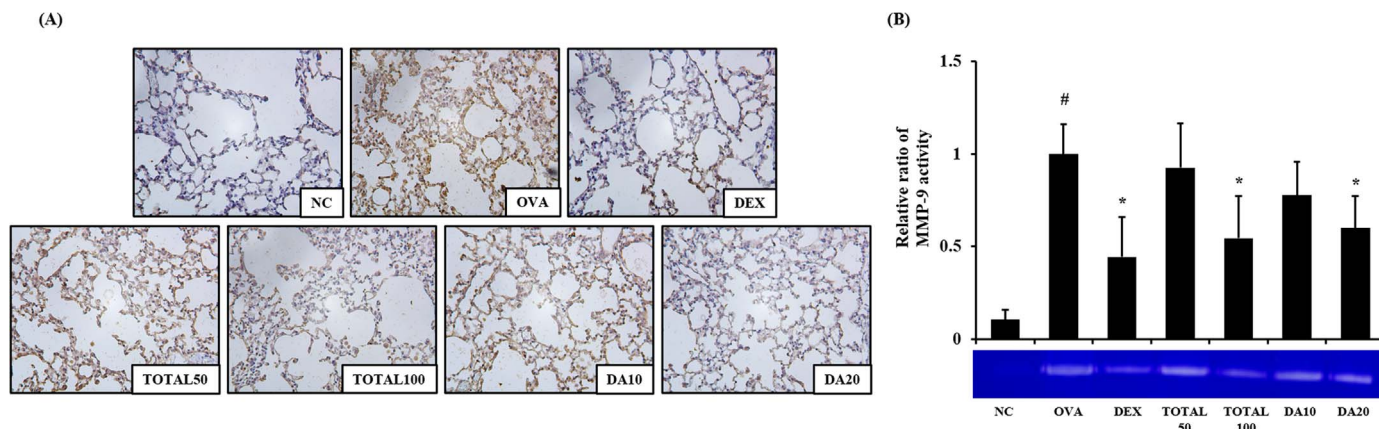


Fig. 5. *Artemisia argyi* inhibited the activity and expression of MMP-9 in lung tissue. (A) Immunohistochemical stain of MMP-9 in lung tissue, (B) MMP-9 activity using zymography. NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. * $P < 0.05$ vs NC; * $P < 0.05$ vs OVA.

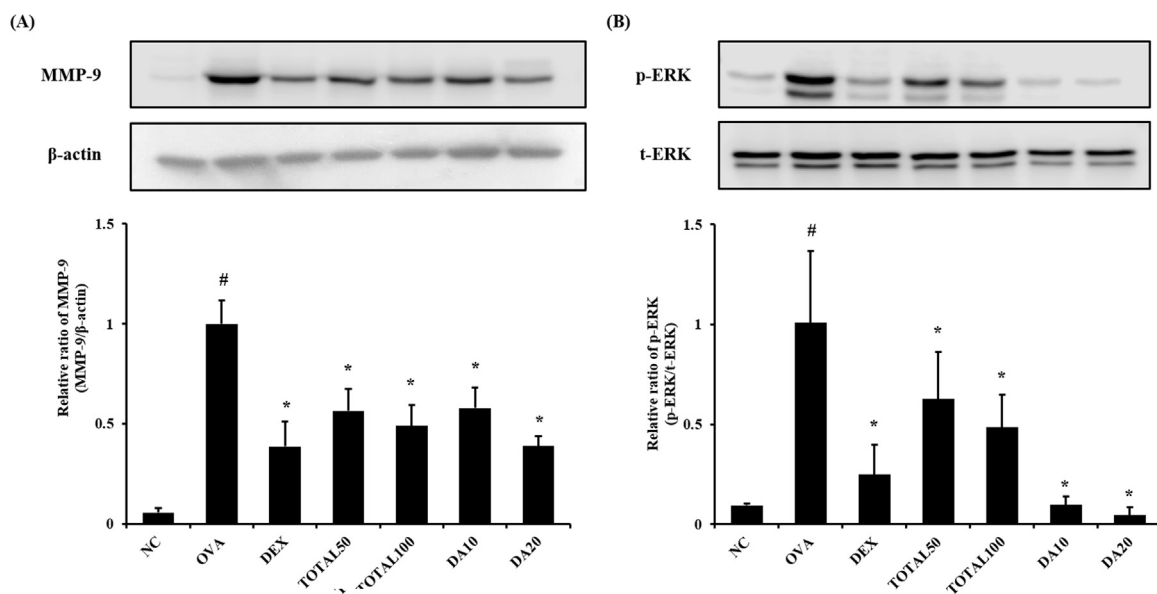


Fig. 6. *Artemisia argyi* decreased MMP-9 and phosphorylation of Erk expression on the protein levels. (A) protein level of MMP-9 expression, (b) protein level of phosphorylation of Erk. NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. # P < 0.05 vs NC; * P < 0.05 vs OVA.

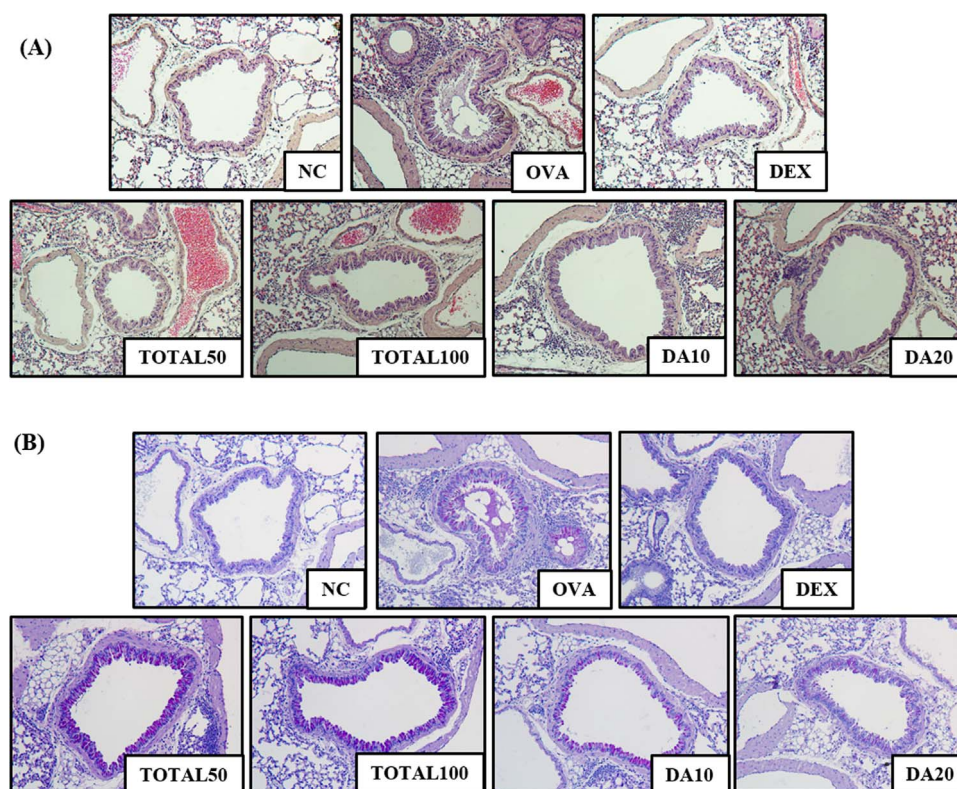


Fig. 7. *Artemisia argyi* reduced inflammatory cell infiltration and mucus production. Hematoxylin and eosin (H & E) staining showed inflammatory accumulation (A) and periodic acid-schiff (PAS) staining showed mucus production (B). NC: no treatment and no OVA challenge; OVA: OVA challenge; DEX: dexamethasone 3 mg/kg per day and OVA challenge; TOTAL50 and 100: 50 and 100 mg/kg of *Artemisia argyi* per day, respectively, and OVA challenge; DA10 and 20: dehydromatricarin A 10 and 20 mg/kg per day, respectively and OVA challenge. The values shown as the means $\pm$ SD. # P < 0.05 vs NC; * P < 0.05 vs OVA.

Walsh et al., 2010). In this study, *Artemisia argyi* and dehydromatricarin A treatment dose-dependently suppressed eosinophil accumulation and elevation of IL-4, IL-5, and IL-13 in BALF that was caused by OVA challenge. In addition, these reductions were accompanied by AHR decline. These results indicate that *Artemisia argyi* may attenuate allergic airway inflammation in OVA-challenged asthmatic mice. This was supported by histological examination. *Artemisia argyi*

treatment alleviated the airway inflammation and mucus hypersecretion in asthmatic animals.

Erk has been implicated in the regulation of cytokines, such as IL-5, that are associated with IgE switching and eosinophil activation (Deng et al., 2013; Wei et al., 2013; Lee et al., 2012a, 2012b). Erk can induce physiological alterations via its phosphorylation, which is caused by various stimuli. Particularly, during an inflammatory process, Erk is

phosphorylated by harmful stimuli, including allergens, chemicals, cytokines, and chemokines, to produce various inflammatory mediators (Wei et al., 2013). MMP-9 is also produced via Erk phosphorylation and is secreted in inflammatory cells (Cheng et al., 2016). MMP-9 is considered a crucial mediator in asthma, and proteinases are expressed in lung and BALF from patients with asthma (Lee et al., 2012a, 2012b; Shin et al., 2015). MMP-9 not only destroys normal alveolar structure, but also aggravates airway inflammation through the degradation of collagen (Cheng et al., 2016). Therefore, down-regulation of Erk/MMP-9 is considered a very important strategy in controlling asthma. In this study, *Artemisia argyi* and dehydromatricarin A treated animals exhibited significant declines of Erk phosphorylation and MMP-9 expression in comparison to those in asthmatic animals, which were detected in dose-dependent manner. These reductions were consistent with the observations of IHC for MMP-9 expression. This suggests that the efficacy of *Artemisia argyi* against allergic responses is closely associated with the downregulation of Erk/MMP-9 signaling in asthmatic animals.

Artemisia argyi is traditionally used as an herbal remedy in Korea, China, and Japan. It has been used to treat sedation, convulsion, and paralysis for a long time in folk medicine, and is prescribed for chronic hepatitis, jaundice, discharge of phlegm, abdominal pain, and respiratory inflammation in oriental medicine (Kim et al., 2015a, 2015b; Yun et al., 2016). Scientific reports have demonstrated that *Artemisia argyi* has various pharmacological activities, including anti-ulcer, anti-inflammation, antioxidant, antibacterial, and immune modulatory effects (Lee et al., 2015; Yao et al., 2016). Therefore, our results strongly corroborate these reports.

In conclusion, *Artemisia argyi* suppressed the elevation of allergic responses, including eosinophilia, increased AHR, production of Th2 cytokines and IgE, and mucus overproduction, in OVA-challenged asthmatic animals. These effects are accompanied with declining MMP-9 expression and Erk phosphorylation in asthmatic animals. Therefore, we suggest that *Artemisia argyi* has a potent agent for treating allergic lung inflammatory diseases.

Conflict of interest

No competing financial interests exist.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jep.2017.07.033.

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